# MA2508 中英双语讲课稿

MA2508 Bilingual Lecture Script

三小时 Tutorial，配合第 1–10 题使用

Three-hour tutorial accompanying Problems 1–10

**中文：**中文与英文逐段对应。三小时安排按每段选择一种语言主讲计算；另一种语言可用于备课或解释关键词。数学板书由两种语言共用。

**English:** Chinese and English are paired paragraph by paragraph. The three-hour schedule assumes that each passage is delivered in one chosen language; use the other for preparation or to clarify key terms. The mathematical board work is shared by both versions.

**中文：**本文中的普通段落是可直接使用的讲解口吻；带有【课堂动作】、【板书】、【提问与预期回答】的内容供讲者查看，不需逐字朗读。每段的时间包含独立练习、板书和互动，不是连续朗读的时长。英文术语可保留在板书中，方便同学与题目和教材对应。

**English:** The ordinary paragraphs are ready-to-use spoken explanations. The sections labelled Classroom action, Board, and Question and expected answer are prompts for the instructor, rather than text to read aloud. Each time slot includes independent work, board writing, and interaction; it is not the duration of continuous reading. Keep the English mathematical terms on the board so that students can connect them with their question sheet and textbook.

## 0–5 分钟：开场与第 1 题 / Opening and Problem 1

**【课堂动作 / Classroom action】**

**中文：**开场约 1 分钟；独立尝试 1 分钟；讲解 2 分钟；核对及过渡 1 分钟。

**English:** Allow about 1 minute for the opening, 1 minute for an independent attempt, 2 minutes for explanation, and 1 minute for checking and transition.

**中文：**大家好，今天我们用十道题一起复习多变量微积分。题目已经按综合难度排列，我们先从短的求导题开始，再到极限、可微和最值。大家不用一开始就把所有公式写满一页；我们每遇到一类题，先判断它在问什么，再选方法。

**English:** Hello everyone. Today we will review multivariable calculus through ten problems. They are arranged by overall difficulty, beginning with short differentiation questions and progressing to limits, differentiability, and extrema. You do not need to start by filling a page with every formula you remember. For each type of problem, first identify what it asks, and then choose the method.

**中文：**做题时我会留一些时间让大家自己试。先写自己的第一步，即使没做完，也比只看完整答案更容易发现自己卡在哪里。我们讲完第五题后休息十分钟。

**English:** I will give you time to try the problems yourselves. Write down your own first step.

Even if you do not finish, this helps you identify where you are stuck more effectively than simply reading a complete solution. We will take a ten-minute break after Problem 5.

**中文：** 现在看第一题。这里给的 $\phi$ 没有具体表达式，不过题目只要求证明一个恒等式，我们可以把 $\phi'$ 留在答案里面。

**English:** Let us look at Problem 1. We are not given an explicit formula for $\phi$, but we only need to prove an identity, so we can leave $\phi'$ in our working.

**中文：** 先把里面重复出现的表达式记成 $q = x^2 - y^2$。对 $x$ 求偏导时，$y$ 固定，所以最前面的 $y$ 导数是零，而后面复合函数给出 $2x\phi'(q)$。对 $y$ 求偏导时，前面的 $y$ 导数是 1，后面是 $-2y\phi'(q)$。

**English:** Write the repeated inner expression as $q = x^2 - y^2$. When differentiating with respect to $x$, hold $y$ fixed: the explicit term $y$ has derivative zero, and the composite term gives $2x\phi'(q)$. When differentiating with respect to $y$, the explicit term contributes 1, and the composite term gives $-2y\phi'(q)$.

**【板书 / Board】**

$$z_x = 2x\phi'(q), \qquad z_y = 1 - 2y\phi'(q).$$

**【提问与预期回答 / Question and expected answer】**

**中文：** "$z_y$ 中这个 1 从哪里来?" 预期回答：从显式项 $y$ 的导数来。

**English:** Where does the 1 in $z_y$ come from? Expected answer: it comes from differentiating the explicit term $y$.

**中文：** 最后别急着继续求导。题目要求的是 $yz_x + xz_y$，我们直接代入。两项 $2xy\phi'(q)$ 正好抵消，剩下 $x$，就证明完成了。

**English:** There is no need to differentiate any further. The question asks for $yz_x + xz_y$, so substitute the two derivatives directly. The two terms involving $2xy\phi'(q)$ cancel, leaving $x$. That completes the proof.

**【板书 / Board】**

$$yz_x + xz_y = 2xy\phi'(q) + x - 2xy\phi'(q) = x.$$

**中文：** 第一题想提醒大家的是，复合项要用 chain rule，但式子中显式出现的变量也要单独处理。接下来我们看两道判断题，练习如何用一个合适的性质把理由说清楚。

**English:** The point of this problem is that the composite term needs the chain rule, while explicitly occurring variables must also be differentiated. Next, we will look at two true-or-false statements and practise giving a clear justification using an appropriate property.

## 5–13 分钟：第 2 题 / Problem 2

**【课堂动作 / Classroom action】**

**中文：** 先给 2 分钟独立判断并写理由，再讲解 5 分钟，最后核对 1 分钟。不要在尝试前公布 True / False。

**English:** Give students 2 minutes to decide and write reasons, explain for 5 minutes, and check for 1 minute. Do not reveal True or False before their attempt.

**中文：** 这一题请大家先不要算一大串坐标。第一小问中，点积为零告诉我们什么？叉积为零又告诉我们什么？注意，题目没有说两个向量非零。

**English:** For this problem, do not start with a long coordinate calculation. In part (a), what does a zero dot product tell us? What does a zero cross product tell us? Notice that the question does not assume that both vectors are nonzero.

**【提问与预期回答 / Question and expected answer】**

**中文：** “如果两个向量都非零，点积为零、叉积为零分别意味着什么?” 预期回答：垂直、平行。

**English:** If both vectors are nonzero, what do a zero dot product and a zero cross product mean, respectively? Expected answer: perpendicular and parallel.

**中文：** 如果两个向量都非零，它们不可能同时平行又垂直。因此至少有一个向量必须为零，这个判断是对的。如果大家想写一个直接包含零向量的代数证明，可以使用这个恒等式。

**English:** Two nonzero vectors cannot be both parallel and perpendicular. Therefore at least one vector must be zero, so the statement is true. If you want an algebraic proof that includes zero vectors directly, use this identity.

**【板书 / Board】**

$$\|\mathbf{u}\times\mathbf{v}\|^2+(\mathbf{u}\cdot\mathbf{v})^2=\|\mathbf{u}\|^2\|\mathbf{v}\|^2.$$

**中文：** 左边两项在题设下都为零，右边的两个非负数之积只能为零，所以其中一个向量的长度为零。这里用长度判断零向量，就不需要讨论零向量的夹角。

**English:** Both terms on the left are zero under the assumptions. The product of the two nonnegative numbers on the right must therefore be zero, so one vector has zero length. Using lengths avoids discussing the angle of a zero vector.

**中文：** 第二小问问的是：有没有一个函数，同时拥有题目给出的这两个一阶偏导？题目还给了连续二阶偏导这个条件，这是关键。它允许我们使用 mixed partial derivatives 相等的结论。

**English:** Part (b) asks whether one function can have both of the proposed first partial derivatives. The assumption of continuous second-order partial derivatives is crucial: it allows us to use equality of the mixed partial derivatives.

**中文：** 我们从 $f_x=x-y^2$ 出发，对 $y$ 求导，得到 $-2y$；从 $f_y=x+y^2$ 出发，对 $x$ 求导，得到 1。假如这样的函数真的存在，这两个结果应该在每一点相同。但在原点，一个是零，一个是一，已经矛盾。

**English:** Differentiate $f_x = x - y^2$ with respect to $y$ to obtain $-2y$, and differentiate $f_y = x + y^2$ with respect to $x$ to obtain 1. If such a function existed, these would agree at every point. At the origin, however, one is zero and the other is one. We already have a contradiction.

**【板书 / Board】**

$$\partial_y f_x = -2y, \qquad \partial_x f_y = 1, \qquad (0,0):\ 0 \neq 1.$$

**【提问与预期回答 / Question and expected answer】**

**中文：**“在 $y = -1/2$ 时它们相等，能不能证明函数存在?”预期回答：不能，题目要求在整个平面上成立。

**English:** They agree when $y = -1/2$. Does that establish the existence of the function? Expected answer: no; the conditions must hold throughout the plane.

**中文：**因此第二小问是错的。判断题的答案不是写完 True 或 False 就结束了，还要指出支持判断的性质或反例。我们刚刚用到了向量，下一题把向量放进一条空间曲线。

**English:** Thus part (b) is false. A true-or-false answer is not complete with the label alone: include the property or counterexample supporting it. We have just worked with vectors; in the next problem, they describe a space curve.

## 13–30 分钟：第 3 题 / Problem 3

**【课堂动作 / Classroom action】**

**中文：**尝试 4 分钟：先求 $\mathbf{r}'$ 和速度，再尝试三个小问。讲解 10 分钟，核对 3 分钟。

**English:** Allow 4 minutes to find $\mathbf{r}'$ and the speed and then attempt the three parts, 10 minutes for explanation, and 3 minutes for checking.

**中文：**这道题有三个小问，但我们不会分别从头算三次。它们都需要 $\mathbf{r}'(t)$，所以先把导数和速度放在板书的一边，后面一直复用。

**English:** There are three parts, but we will not start the calculation three times. All three use $\mathbf{r}'(t)$, so keep the derivatives and speed on one side of the board and reuse them.

**中文：**开始之前，我们要知道点 $P = (1,0,1)$ 对应哪个参数。看第三坐标最方便：$e^t = 1$，所以 $t = 0$。得到参数以后再求这个点的切向量和曲率。

**English:** First determine which parameter corresponds to $P = (1,0,1)$. The third coordinate gives the quickest route: $e^t = 1$, so $t = 0$. Then evaluate the tangent vector and curvature at that parameter.

**【板书 / Board】**

$$\mathbf{r}'(t) = e^t(\cos t - \sin t, \sin t + \cos t, 1),$$
$$\mathbf{r}''(t) = e^t(-2\sin t, 2\cos t, 1).$$

**中文：** 请大家看前两个分量的平方和。展开后，交叉项抵消，$\sin^2 t + \cos^2 t$ 出现两次，再加第三分量的平方，结果是 $3e^{2t}$。因此速度是 $\sqrt{3}\,e^t$。这里 $e^t$ 始终为正，所以开根号不用再分正负情况。

**English:** Look at the sum of the squares of the first two components. On expansion, the cross terms cancel, and $\sin^2 t + \cos^2 t$ occurs twice. Including the third component gives $3e^{2t}$. Hence the speed is $\sqrt{3}\,e^t$. Since $e^t$ is positive, taking the square root requires no sign cases.

**【板书 / Board】**

$$\|\mathbf{r}'(t)\| = \sqrt{3}\,e^t.$$

**中文：** 第一小问，unit tangent vector 就是导向量除以它的长度。零点的导向量是 $(1,1,1)$，长度是 $\sqrt{3}$，所以结果是 $(1,1,1)/\sqrt{3}$。算完可以检查它的长度是不是一。

**English:** In part (a), the unit tangent vector is the derivative vector divided by its length. At $t = 0$, the derivative is $(1,1,1)$ and its length is $\sqrt{3}$, giving $(1,1,1)/\sqrt{3}$. Check that the result has length one.

**中文：** 第二小问求 arc length，积分的对象是速度。我们积分的是 $\sqrt{3}\,e^t$，不是位置向量的长度，也不是单独某一个坐标的导数。从零积到 $\ln 2$，得到 $\sqrt{3}(2-1) = \sqrt{3}$。

**English:** Part (b) asks for arc length, so we integrate the speed. The integrand is $\sqrt{3}\,e^t$, rather than the length of the position vector or the derivative of a single coordinate. Integrating from zero to $\ln 2$ gives $\sqrt{3}(2-1) = \sqrt{3}$.

**【提问与预期回答 / Question and expected answer】**

**中文：** “为什么积分上限 $\ln 2$ 很方便?” 预期回答：$e^{\ln 2} = 2$。

**English:** Why is the upper limit $\ln 2$ convenient? Expected answer: $e^{\ln 2} = 2$.

**中文：** 第三小问求 curvature。我们使用叉积公式，分母是速度的三次方。这一题只问 $P$ 处的曲率，所以没必要把任意参数的叉积完整展开；直接算 $t = 0$ 就可以了。

**English:** For part (c), use the cross-product formula for curvature, whose denominator is the cube of the speed. We only need curvature at $P$, so there is no reason to expand the cross product for general $t$. Evaluate it at $t = 0$.

**【板书 / Board】**

$$\mathbf{r}'(0) = (1,1,1), \quad \mathbf{r}''(0) = (0,2,1), \quad \mathbf{r}'(0) \times \mathbf{r}''(0) = (-1,-1,2).$$

**中文：** 叉积长度是 $\sqrt{6}$，速度是 $\sqrt{3}$，所以曲率是 $\sqrt{6}/(\sqrt{3})^3 = \sqrt{2}/3$。算叉积时请检查第二个分量的符号，最后也检查分母有没有写成三次方。

**English:** The cross product has length $\sqrt{6}$, and the speed is $\sqrt{3}$. Thus the curvature is $\sqrt{6}/(\sqrt{3})^3 = \sqrt{2}/3$. Check the sign of the second cross-product component, and make sure that the denominator uses the third power.

**【板书 / Board】**

$$\mathbf{T}(0) = \frac{(1,1,1)}{\sqrt{3}}, \qquad L = \sqrt{3}, \qquad \kappa(0) = \frac{\sqrt{2}}{3}.$$

**中文：** 现在请大家用这三个结果核对自己的步骤。我们从“沿曲线的方向”过渡到“沿平面中指定方向的变化率”，以及曲面的切平面。

**English:** Use these three results to check your work. We will now move from the direction along a curve to rates of change in a specified direction, and to tangent planes of surfaces.

## 30–52 分钟：第 4 题 / Problem 4

**【课堂动作 / Classroom action】**

**中文：** 独立尝试 5 分钟；分三小问讲解约 14 分钟；核对 3 分钟。板书把三个梯度分别写在对应小问下，不混用函数记号。

**English:** Allow 5 minutes for independent attempts, about 14 minutes to explain the three parts, and 3 minutes to check. Write each gradient beneath its corresponding part, keeping the function names distinct.

**中文：** 第一小问问 directional derivative。我们的公式是梯度与单位方向向量的点积。请注意，题目给的 $(1,1)$ 长度是 $\sqrt{2}$，还不是单位向量。

**English:** Part (a) asks for a directional derivative. The formula is the dot product of the gradient with a unit direction vector. The given vector $(1,1)$ has length $\sqrt{2}$, so it is not yet a unit vector.

**中文：** 先单位化，再求梯度。$h_x = -y\sin x$，$h_y = \cos x + 2y$，所以在 $(0,1)$，梯度是 $(0,3)$。与 $(1,1)/\sqrt{2}$ 做点积，得到 $3/\sqrt{2}$。

**English:** Normalize the direction, then find the gradient. We have $h_x = -y\sin x$ and $h_y = \cos x + 2y$, so at $(0,1)$ the gradient is $(0,3)$. Taking its dot product with $(1,1)/\sqrt{2}$ gives $3/\sqrt{2}$.

**【板书 / Board】**

$$\widehat{\mathbf{a}} = \frac{(1,1)}{\sqrt{2}}, \quad \nabla h(0,1) = (0,3), \quad D_{\widehat{\mathbf{a}}}h = \frac{3}{\sqrt{2}}.$$

**【提问与预期回答 / Question and expected answer】**

**中文：** “如果直接用 $(1,1)$ 做点积，会得到什么？为什么不符合单位方向导数的定义？”预期回答：会得到 3，忽略了方向向量的长度。

**English:** What would you obtain by taking the dot product directly with $(1,1)$, and why would that miss the unit-direction definition? Expected answer: 3; it ignores the length of the direction vector.

**中文：** 第二小问，题目给了曲面和切点。我们把曲面写成 $F(x,y,z) = 0$。只要完整梯度不为零，它就是切平面的法向量。

**English:** In part (b), both the surface and the point of tangency are given. Write the surface as $F(x, y, z) = 0$. If the full gradient is nonzero, it provides a normal vector to the tangent plane.

**中文：** 这里 $F = z - e^z + 2xy - 3$，梯度是 $(2y, 2x, 1 - e^z)$。代入 $(1, 2, 0)$ 后是 $(4, 2, 0)$。法向量第三分量为零，只表示切平面方程没有 $z$ 项，不代表切平面不存在。

**English:** Here $F = z - e^z + 2xy - 3$, with gradient $(2y, 2x, 1 - e^z)$. At $(1, 2, 0)$, this becomes $(4, 2, 0)$. A zero third component simply means that the tangent-plane equation has no $z$ term. It does not mean that the tangent plane fails to exist.

**【板书 / Board】**

$$\nabla F(1, 2, 0) = (4, 2, 0), \qquad 4(x - 1) + 2(y - 2) = 0, \qquad 2x + y = 4.$$

**中文：** 有同学可能想把它改写成 $z = z(x, y)$，然后算 $z_x, z_y$。这个点上 $F_z = 0$，那条路线不合适。我们直接使用水平集的完整梯度，已经足够了。

**English:** You might try to write the surface as $z = z(x, y)$ and compute $z_x$ and $z_y$. Since $F_z = 0$ at this point, that route is unsuitable here. The full gradient of the level surface is all we need.

**中文：** 第三小问不一样：现在切点没有给出来，只知道切平面要与一个指定平面平行。我们先从平行条件找切点，再写平面方程。

**English:** Part (c) is different: the point of tangency is unknown, but the tangent plane must be parallel to a specified plane. First use parallelism to find the points, then write the plane equations.

**中文：** 给定平面的法向量是 $(1, 3, 5)$。椭球的法向量是梯度 $(2x, 4y, 6z)$，所以设它们之间存在比例系数 $\lambda$。注意，是梯度与给定法向量成比例，不是坐标直接与法向量成比例。

**English:** The given plane has normal vector $(1, 3, 5)$. The ellipsoid has gradient $(2x, 4y, 6z)$, so set these vectors proportional using $\lambda$. The gradient, rather than the coordinates themselves, must be proportional to the given normal.

**【板书 / Board】**

$$(2x, 4y, 6z) = \lambda(1, 3, 5) \quad \Rightarrow \quad x = \frac{\lambda}{2},\ y = \frac{3\lambda}{4},\ z = \frac{5\lambda}{6}.$$

**中文：** 把这三个表达式代回椭球，得到 $(83/24)\lambda^2 = 498$，因此 $\lambda = \pm 12$。所以有两个切点，分别是 $(6, 9, 10)$ 和它的相反点。

**English:** Substituting the three coordinate expressions into the ellipsoid gives $(83/24)\lambda^2 = 498$, hence $\lambda = \pm 12$. There are two points of tangency: $(6, 9, 10)$ and its negative.

**中文：** 最后用法向量 $(1, 3, 5)$ 写平面。正切点代入 $x + 3y + 5z$ 得到 $83$，负切点得到 $-83$，两个切平面就是 $x + 3y + 5z = \pm 83$。

**English:** Finally use normal vector $(1, 3, 5)$ to write each plane. At the positive point, $x+3y+5z = 83$; at the negative point, it is $-83$. Therefore the tangent planes are $x + 3y + 5z = \pm 83$.

**【提问与预期回答 / Question and expected answer】**

**中文：**“为什么不是等于题目中的 7?”预期回答：平行只确定法向量方向，不确定平面的位置。

**English:** Why is the constant not the 7 given in the question? Expected answer: parallelism fixes the normal direction, not the position of the plane.

**中文：**三个小问的共同工具是 gradient，但使用目的不同：一个算变化率，一个在给定点找法向量，一个通过法向量反求切点。下面进入极限，方法的选择会更重要。

**English:** The gradient is the common tool in these three parts, but serves three purposes: calculating a rate of change, finding a normal at a specified point, and recovering points of tangency from a normal direction. Next we turn to limits, where choosing the method matters even more.

## 52–77 分钟：第 5 题 / Problem 5

**【课堂动作 / Classroom action】**

**中文：**先给 6 分钟尝试。讲解约 16 分钟：有理化约 4 分钟，夹逼约 5 分钟，路径反例约 7 分钟。核对及休息安排 3 分钟。

**English:** Allow 6 minutes for attempts and about 16 minutes for explanation: 4 for rationalization, 5 for the squeeze argument, and 7 for the path counterexample. Use 3 minutes for checking and announcing the break.

**中文：**这三小问请先想一想：表达式有没有适合的代换？如果你觉得极限存在，能不能给出覆盖所有方向的理由？如果你觉得不存在，能不能找出两条合法路径，得到不同的结果？

**English:** For each part, ask: is there a useful substitution? If you think the limit exists, can you justify it for every approach direction? If you think it does not exist, can you find two valid paths giving different limits?

**中文：**第一小问，式子中的变量总是以 $xy$ 的形式出现，所以令 $s = xy$。接近原点时 $s$ 趋于零，我们把问题化成一个一元极限。

**English:** In part (a), the variables occur together as $xy$, so put $s = xy$. As we approach the origin, $s$ tends to zero, reducing the problem to a one-variable limit.

**中文：**分母是根号减一，先有理化。乘上根号加一以后，分母变成 $1 - e^s$。这里特别容易丢负号，因为常用的指数极限是 $(e^s - 1)/s \to 1$。

**English:** The denominator is a square root minus one, so rationalize it. Multiplying by the conjugate gives denominator $1 - e^s$. Watch the minus sign: the familiar exponential limit is $(e^s - 1)/s \to 1$.

**【板书 / Board】**

$$\frac{s}{\sqrt{2 - e^s} - 1} = \frac{s(\sqrt{2 - e^s} + 1)}{1 - e^s} = -\frac{s}{e^s - 1}(\sqrt{2 - e^s} + 1).$$

**中文：** 第一部分趋于负一，括号趋于二，因此极限是负二。原表达式在 $xy = 0$ 上没有定义，但题目是沿它的定义域求极限；这不妨碍我们得到这个结果。

**English:** The first factor tends to negative one, and the bracket tends to two, giving limit negative two. The original expression is undefined on $xy = 0$, but the limit is taken through its domain, so this causes no difficulty.

**中文：** 第二小问，我们预期结果是零。试 $x = 0$、$y = 0$ 都得到零，但这还不是证明。我们要找一个对所有方向都有效的上界。

**English:** In part (b), we expect zero. The paths $x = 0$ and $y = 0$ both give zero, but that is not a proof. We need a bound valid in every direction.

**中文：** 设 $r = \sqrt{x^2 + y^2}$。因为 $|x|, |y|$ 都不超过 $r$，所以 $|x|^3 \le rx^2$，$|y|^3 \le ry^2$。把分子绝对值用三角不等式估计，得到整个表达式的绝对值不超过 $r$，而 $r \to 0$。

**English:** Let $r = \sqrt{x^2 + y^2}$. Since $|x|$ and $|y|$ are at most $r$, we have $|x|^3 \le rx^2$ and $|y|^3 \le ry^2$. Apply the triangle inequality to the numerator: the absolute value of the whole expression is at most $r$, which tends to zero.

**【板书 / Board】**

$$\left|\frac{x^3 - y^3}{x^2 + y^2}\right| \le \frac{|x|^3 + |y|^3}{r^2} \le \frac{r(x^2 + y^2)}{r^2} = r \to 0.$$

**中文：** 这里的重点是这个界没有指定角度，所有接近原点的点都受它控制。用 polar coordinates 也可以，但必须说明角度部分统一有界，不能只固定一个角度就宣布完整极限存在。

**English:** The bound does not select an angle: it controls all points approaching the origin. Polar coordinates also work, provided the angular factor is uniformly bounded. Holding one angle fixed does not prove the full limit.

**中文：** 第三小问，如果大家试了若干直线，很可能都得到零。这时候不能停止。我们看分母 $x^2 + y^6$，把两项平衡起来：令 $x = y^3$，就能让两项具有相同量级。

**English:** For part (c), trying several straight lines may give zero every time. Do not stop there. Balance the two denominator terms $x^2$ and $y^6$: choosing $x = y^3$ makes them comparable.

**【课堂动作 / Classroom action】**

**中文：** 先只板书 $x^2 = y^6$，停顿，让学生说出可选路径；不要立即写出最终值。

**English:** First write only $x^2 = y^6$, then pause and ask students to suggest a path. Do not reveal the final value immediately.

**中文：** 沿 $x = y^3$，分子是 $y^6$，分母是 $2y^6$，结果是 $1/2$。沿 $x = 0$，结果是零。这两条路径都在定义域中趋向原点，却得到不同的极限，所以完整极限不存在。

**English:** Along $x = y^3$, the numerator is $y^6$ and the denominator is $2y^6$, giving $1/2$. Along $x = 0$, the value is zero. Both paths approach the origin through the domain, but their limits differ, so the full limit does not exist.

**【板书 / Board】**

$$x = y^3: \ \frac{xy^3}{x^2+y^6} = \frac{1}{2}, \qquad x = 0: \ \frac{xy^3}{x^2+y^6} = 0.$$

**中文：**我们再回头看直线。沿 $y = mx$，表达式是 $m^3x^2/(1+m^6x^4)$，确实趋于零，竖直线也得到零。这说明即使所有直线都一致，也可能有曲线路径让结果不一样。

**English:** Now return to straight lines. Along $y = mx$, the expression is $m^3x^2/(1+m^6x^4)$, which does tend to zero; the vertical line does too. Thus agreement along every straight line can still coexist with a different limit along a curve.

**【提问与预期回答 / Question and expected answer】**

**中文：**“证明不存在和证明存在，对证据的要求有什么差别?”预期回答：不同路径可以否定；存在需要覆盖所有合法接近方式的论证。

**English:** How does the evidence needed to prove nonexistence differ from the evidence needed to prove existence? Expected answer: disagreeing paths disprove a limit; existence requires an argument covering every valid approach.

**中文：**这一组结果分别是负二、零、不存在。请大家检查自己有没有把“几条路径相同”写成存在性的依据。我们现在休息十分钟，回来后用类似的极限思想检查可微性。

**English:** The answers are negative two, zero, and nonexistence. Check whether you treated agreement along a few paths as proof of existence. We will now take a ten-minute break; afterwards, we will use similar limit ideas to investigate differentiability.

## 77–87 分钟：休息 / Break

**【课堂动作 / Classroom action】**

**中文：**休息十分钟。保留第 5 题的结果和“路径相同不够证明存在”的板书；回来后用它过渡到第 6 题。

**English:** Take a ten-minute break. Keep the results of Problem 5 and the reminder that agreeing path limits do not prove existence on the board, then connect this to Problem 6 when students return.

## 87–102 分钟：第 6 题 / Problem 6

**【课堂动作 / Classroom action】**

**中文：**独立尝试 4 分钟；讲解 9 分钟；核对 2 分钟。板书按三个小问分成三行结论，避免混用定义。

**English:** Allow 4 minutes for independent attempts, 9 minutes for explanation, and 2 minutes for checking. Write the three conclusions separately so that the definitions stay distinct.

**中文：** 现在这道题要判断三件不同的事：continuous、partial derivatives exist、differentiable。大家先不要从前两个小问直接猜第三个。我们每一个都用自己的定义或充分的估计来检查。

**English:** This problem asks about three different properties: continuity, existence of partial derivatives, and differentiability. Do not infer part (c) just from the first two parts. Check each property using its definition or a sufficient estimate.

**中文：** 第一小问，连续要求函数值趋于它在原点的值，也就是零。这里可以用 $2|xy| \le x^2 + y^2$。分子除以距离以后，得到绝对值最多是距离的一半，所以函数趋于零，原点连续。

**English:** For part (a), continuity means that the function tends to its value at the origin, namely zero. Use $2|xy| \le x^2 + y^2$. After division by the distance, the absolute value is at most half the distance, so the function tends to zero and is continuous at the origin.

**【板书 / Board】**

$$|f(x,y)| \le \frac{1}{2}\sqrt{x^2+y^2} \to 0 = f(0,0).$$

**中文：** 第二小问，用偏导定义。固定 $y = 0$，函数沿横轴恒为零，所以 $f_x(0,0) = 0$；固定 $x = 0$，沿纵轴也恒为零，所以 $f_y(0,0) = 0$。这里不要把原点塞进非原点的公式，因为原函数在那里本来就是另行定义的。

**English:** For part (b), use the definitions of the partial derivatives. On $y = 0$, the function is identically zero, giving $f_x(0,0) = 0$. On $x = 0$, it is also identically zero, giving $f_y(0,0) = 0$. Do not substitute the origin into a formula derived away from it, because the function is separately defined there.

**中文：** 第三小问，如果函数可微，它的线性近似应该由这两个偏导决定。由于两个偏导都是零，候选线性映射就是零。我们要检查函数与零线性映射之间的误差，除以距离之后，是否还趋于零。

**English:** For part (c), if the function were differentiable, its linear approximation would be determined by these partial derivatives. Both are zero, so the candidate linear map is zero. Check whether the error from this zero linear map, divided by the distance, tends to zero.

**【板书 / Board】**

$$\frac{f(h,k) - f(0,0) - f_x(0,0)h - f_y(0,0)k}{\sqrt{h^2+k^2}} = \frac{hk}{h^2+k^2}.$$

**中文：** 沿 $k = h$，右边恒等于 $1/2$，所以不趋于零，函数在原点不可微。

**English:** Along $k = h$, the quotient is constantly $1/2$, so it does not tend to zero. Therefore the function is not differentiable at the origin.

**【提问与预期回答 / Question and expected answer】**

**中文：** “可是沿 $y = x$，函数本身还是趋于零，这不矛盾吗?” 预期回答：不矛盾，连续看函数本身，可微看误差相对于距离的大小。

**English:** But the function itself still tends to zero along $y = x$. Is that a contradiction? Expected

answer: no; continuity concerns the function value, while differentiability concerns the error relative to the distance.

**中文：** 所以这道题的三个结论是：连续，偏导存在且都为零，不可微。大家要记住这三个概念之间的区别。下一题回到计算，但复合关系更复杂，我们要控制展开的顺序。

**English:** The three conclusions are: continuous; both partial derivatives exist and equal zero; not differentiable. Keep these properties distinct. The next problem returns to calculation, but with a more complicated composite dependence, so the order of expansion becomes important.

## 102–122 分钟：第 7 题 / Problem 7

**【课堂动作 / Classroom action】**

**中文：** 独立尝试 5 分钟；分层讲解 12 分钟；核对 3 分钟。尽量保留中间式，不边口述边连续擦除。

**English:** Allow 5 minutes for an independent attempt, 12 minutes for a layered explanation, and 3 minutes for checking. Keep intermediate expressions visible instead of repeatedly erasing while speaking.

**中文：** 这道题的外层是 $w = f(u, v)$，内层的 $u, v$ 又依赖 $x, y, z$。如果把每一个指数都展开在同一行，很容易漏项。因此我们先写依赖关系，再列内层的导数。

**English:** The outer function is $w = f(u, v)$, while $u$ and $v$ depend on $x, y, z$. Expanding every exponential in one line makes it easy to omit terms. First record the dependence, then list the inner derivatives.

**中文：** 为了少写一些符号，暂时把 $e^{xyz}$ 记成 $E$。我们仍然知道 $E$ 随 $z$ 变化，记成一个字母不代表它是常数。

**English:** To shorten the notation, temporarily write $E = e^{xyz}$. It still depends on $z$: replacing it by a letter does not make it a constant.

**【板书 / Board】**

$$u_y = z, \quad u_z = y, \quad v_y = xzE, \quad v_z = xyE.$$

**中文：** 第一阶很直接：$w_y = f_u u_y + f_v v_y$，所以是 $zf_u + xzEf_v$。请把它留在板书上，因为第二阶就是对这整个式子再求一次 $z$ 偏导。

**English:** The first derivative is straightforward: $w_y = f_u u_y + f_v v_y = zf_u + xzEf_v$. Keep this expression on the board, because the second derivative is obtained by differentiating this entire expression with respect to $z$.

**中文：** 这里有两个乘积。第一个 $zf_u$ 中，$z$ 会变，$f_u$ 也会变；第二个 $xzEf_v$ 中，前面的系数会变，$f_v$ 也会变。所以不能只把 $f_u$ 改成某个二阶导。

**English:** There are two products. In $zf_u$, both $z$ and $f_u$ vary. In $xzEf_v$, both the coefficient and $f_v$ vary. We cannot merely replace $f_u$ with a second derivative and leave everything else untouched.

**【板书 / Board】**

$$\begin{aligned}\partial_z f_u &= y f_{uu} + xyE f_{uv},\\ \partial_z f_v &= y f_{vu} + xyE f_{vv},\\ \partial_z(xzE) &= xE + x^2yzE.\end{aligned}$$

**中文：** 先把这三条单独算清楚，再放回乘积法则中。第一项产生 $f_u$，以及 $z$ 乘上 $\partial_z f_u$。第二项产生 $\partial_z(xzE)$ 乘上 $f_v$，以及 $xzE$ 乘上 $\partial_z f_v$。

**English:** Calculate these three derivatives separately, then substitute them into the product rule. The first product contributes $f_u$ and $z$ times $\partial_z f_u$. The second contributes $\partial_z(xzE)$ times $f_v$, and $xzE$ times $\partial_z f_v$.

**【板书 / Board】**

$$\begin{aligned}\partial_z(w_y) = f_u &+ z(y f_{uu} + xyE f_{uv})\\ &+ (xE + x^2yzE) f_v + xzE(y f_{vu} + xyE f_{vv}).\end{aligned}$$

**中文：** 题目给了连续二阶偏导，所以 $f_{uv} = f_{vu}$。到这一步再合并，得到最后的五种项。

**English:** The second-order partial derivatives are continuous, so $f_{uv} = f_{vu}$. Only now combine the terms to obtain the final five types of contribution.

**【板书 / Board】**

$$\begin{aligned}\partial_z(w_y) = yz f_{uu} + 2xyzE f_{uv} + x^2yzE^2 f_{vv}\\ + f_u + x(1 + xyz)E f_v.\end{aligned}$$

**中文：** 现在我们不重新算一遍，而是做一个局部检查。$f_{vv}$ 的系数应当来自 $v_y v_z$。两个系数相乘是 $(xzE)(xyE) = x^2yzE^2$，只有一个 $z$，不是 $z^2$。

**English:** Rather than repeat the entire calculation, check one coefficient independently. The coefficient of $f_{vv}$ should be $v_y v_z$. Their product is $(xzE)(xyE) = x^2yzE^2$, with one factor of $z$, not $z^2$.

**【提问与预期回答 / Question and expected answer】**

**中文：** “最后的 $f_u$ 是从哪个步骤出现的?” 预期回答：对第一个乘积 $zf_u$ 中的 $z$ 求导。

**English:** Where does the final $f_u$ term come from? Expected answer: from differentiating the factor $z$ in the first product $zf_u$.

**中文：** 这题的计算框架比最终长公式更值得记住：列内层导数，做一阶链式法则，再对每个乘积完整求导，最后合并。接下来换一道短的概念题，继续检查路径检验的局限。

**English:** Remember the structure of the calculation: list inner derivatives, apply the first-order

chain rule, differentiate every product completely, and combine terms last. Next comes a short conceptual problem revisiting the limitations of path tests.

## 122–127 分钟：第 8 题 / Problem 8

**【课堂动作 / Classroom action】**

**中文：** 尝试 1 分钟，讲解 3 分钟，核对 1 分钟。先画区域，不逐点列函数值。

**English:** Allow 1 minute for an attempt, 3 minutes for explanation, and 1 minute for checking. Sketch the region first, rather than listing values point by point.

**中文：** 这道分段函数什么时候等于一？只有在 $y = 0$ 上方、$y = x^6$ 下方这条狭窄区域内。其余位置都等于零。

**English:** When is this piecewise function equal to one? Only in the narrow region above $y = 0$ and below $y = x^6$. Everywhere else it is zero.

**中文：** 对于第一小问给出的路径，如果 $m \le 0$，那么 $y \le 0$，函数一路都是零。如果 $m > 0$，我们看高度比例 $y/x^6 = m|x|^{a-6}$。由于 $a < 6$，这个比例趋向正无穷，所以路径靠近原点时反而在这条狭窄区域的上方，函数最终也是零。

**English:** For the paths in part (a), if $m \le 0$, then $y \le 0$, and the function is zero throughout. If $m > 0$, examine the ratio $y/x^6 = m|x|^{a-6}$. Since $a < 6$, this ratio tends to positive infinity. Near the origin, the path eventually lies above the narrow region, so the function is eventually zero again.

**【板书 / Board】**

$$m > 0: \quad \frac{y}{x^6} = m|x|^{a-6} \to +\infty.$$

**中文：** 现在我们换一条路径，沿 $y = x^6/2$，每个非原点的点都在狭窄区域内部，函数恒为一。沿 $y = 0$，函数恒为零，所以完整极限不存在，原点不连续。

**English:** Choose a different path, $y = x^6/2$. Every non-origin point on it lies inside the narrow region, so the function is constantly one. On $y = 0$ it is constantly zero. Hence the full limit does not exist, and the function is discontinuous at the origin.

**【提问与预期回答 / Question and expected answer】**

**中文：** “这条反例为什么没有包含在第一小问的路径族里?” 预期回答：它的指数是 6，而第一小问只覆盖 $a < 6$。

**English:** Why was this counterexample not covered by the path family in part (a)? Expected answer: its exponent is 6, whereas part (a) only covers $a < 6$.

**中文：** 这一题计算不长，难的是识别题目没有检查到的路径。下面两题是最值题，我们会把 “有没有漏掉情况” 作为主要检查点。

**English:** The calculation is short; the challenge is identifying a path that the first test misses. The last two problems concern extrema, where our main checkpoint is whether we have omitted any cases.

## 127–150 分钟：第 9 题 / Problem 9

**【课堂动作 / Classroom action】**

**中文：**给 6 分钟独立建立并解方程；讲解 14 分钟；核对 3 分钟。若进度紧张，保留乘子法完整分类，缩短最后的不等式验证。

**English:** Give 6 minutes to set up and solve the equations, explain for 14 minutes, and check for 3 minutes. If time is tight, keep the complete multiplier case split and shorten the final inequality verification.

**中文：**第九题在球面上求最值。先说清楚为什么可以找绝对最大最小：球面闭且有界，目标函数连续，所以极值存在。而约束梯度在这个球面上不为零，因此 Lagrange multiplier 的必要条件可以用来找全部极值候选点。

**English:** Problem 9 seeks extrema on a sphere. First justify existence: the sphere is closed and bounded, and the objective is continuous, so absolute extrema exist. The constraint gradient is nonzero on the sphere, so the Lagrange multiplier necessary condition applies to every extremum candidate.

**中文：**我们采用 $\nabla f = \lambda \nabla G$ 的写法，得到这四条方程。

**English:** Using $\nabla f = \lambda \nabla G$ gives these four equations.

**【板书 / Board】**

$$y = 2\lambda x, \quad x = 2\lambda y, \quad 2 = 2\lambda z, \quad x^2 + y^2 + z^2 = 11.$$

**中文：**现在重点来了。把第一条代进第二条，得到 $x(1 - 4\lambda^2) = 0$。请先停一下，不要直接除以 $x$。这个乘积为零，意味着两种可能：$x = 0$，或者 $1 - 4\lambda^2 = 0$。

**English:** Here is the crucial step. Substitute the first equation into the second to obtain $x(1 - 4\lambda^2) = 0$. Pause before dividing by $x$. A zero product gives two possibilities: $x = 0$, or $1 - 4\lambda^2 = 0$.

**【提问与预期回答 / Question and expected answer】**

**中文：**“如果直接除以 $x$，我们会丢掉什么?”预期回答：所有 $x = 0$ 的候选点。

**English:** What would we lose by dividing directly by $x$? Expected answer: all candidates with $x = 0$.

**中文：**先看零分支。$x = 0$ 代进第一条，得到 $y = 0$；球面约束给出 $z = \pm\sqrt{11}$，函数值是 $\pm 2\sqrt{11}$。这两个点都要列在候选表中。它们是不是最终最值，我们等全部算完后再决定。

**English:** Start with the zero branch. Substituting $x = 0$ into the first equation gives $y = 0$, and the sphere constraint gives $z = \pm\sqrt{11}$. The function values are $\pm 2\sqrt{11}$. List both points as candidates; decide whether they are the final extrema only after calculating every branch.

**中文：** 接着看 $\lambda = 1/2$。这时 $y = x$，第三条给出 $z = 2$。约束变成 $2x^2 + 4 = 11$，所以 $x = \pm\sqrt{7/2}$，两个点的 $x, y$ 同号。函数值是 $7/2 + 4 = 15/2$。

**English:** Next take $\lambda = 1/2$. Then $y = x$, and the third equation gives $z = 2$. The constraint becomes $2x^2 + 4 = 11$, so $x = \pm\sqrt{7/2}$. The coordinates $x, y$ have the same sign at these two points. The function value is $7/2 + 4 = 15/2$.

**中文：** 再看 $\lambda = -1/2$。这时 $y = -x$，$z = -2$，同样得到 $x^2 = 7/2$。两个点的 $x, y$ 异号，函数值是 $-7/2 - 4 = -15/2$。

**English:** For $\lambda = -1/2$, we have $y = -x$ and $z = -2$, again with $x^2 = 7/2$. The coordinates $x, y$ now have opposite signs. The function value is $-7/2 - 4 = -15/2$.

**【板书 / Board】**

$a = \sqrt{7/2},$

| 分支 / Case | 候选点 / Candidates | $f$ |
|---|---|---|
| $x = 0$ | $(0, 0, \pm\sqrt{11})$ | $\pm 2\sqrt{11}$ |
| $\lambda = 1/2$ | $(a, a, 2), (-a, -a, 2)$ | $15/2$ |
| $\lambda = -1/2$ | $(a, -a, -2), (-a, a, -2)$ | $-15/2$ |

**中文：** 把表中的值比较，就知道最大值是 $15/2$，最小值是 $-15/2$。请把对应的四个点全部写出来，而不是只写两个数。我们的分类是由一个乘积为零得到的完整分类，所以没有遗漏其他乘子分支。

**English:** Compare the values in the table. The maximum is $15/2$, and the minimum is $-15/2$. Write all four corresponding points, not merely the two values. Our case split exhausts the zero-product equation, so no other multiplier branches have been omitted.

**中文：** 如果还有时间，我们可以用一个不等式检查数值。$|xy| \le (x^2+y^2)/2$，而球面上 $x^2+y^2 = 11-z^2$。把上下界配方，就得到 $f$ 不超过 $15/2$，也不低于 $-15/2$。

**English:** If time permits, verify the values with an inequality. We have $|xy| \le (x^2 + y^2)/2$, and on the sphere $x^2 + y^2 = 11 - z^2$. Completing the square in the upper and lower bounds shows that $f$ is at most $15/2$ and at least $-15/2$.

**【板书 / Board】**

$$-\frac{15}{2} + \frac{(z+2)^2}{2} \le xy + 2z \le \frac{15}{2} - \frac{(z-2)^2}{2}.$$

**中文：** 这个检查与前面的结果相符，但我们仍然保留题目要求的乘子法。最后一题的约束不只是一个曲面，而是一整块有边界的区域，我们要把内点和每段边界分别处理。

**English:** This check agrees with our result, while the multiplier calculation remains the method requested by the question. In the last problem, the domain is an entire region with a boundary, rather than just a surface. We must examine its interior and each boundary component separately.

# 150–175 分钟：第 10 题 / Problem 10

**【课堂动作 / Classroom action】**

**中文：** 尝试 6 分钟：要求先画区域、列出需要检查的部分。讲解 16 分钟，核对 3 分钟。板书三列或三块区域：内点、直径、圆弧。

**English:** Allow 6 minutes for an attempt: ask students to sketch the region and list all components to check. Explain for 16 minutes and check for 3 minutes. Divide the board into interior, diameter, and circular arc.

**中文：** 第十题请大家先画 $D$。这是半径二的上半圆盘，包括里面所有点，也包括横轴上的直径。看到 $x^2+y^2\le 4$，不能只在圆弧上找最值。

**English:** First sketch $D$. It is the upper half-disc of radius two, including all interior points and the diameter on the horizontal axis. The condition $x^2+y^2\le 4$ does not restrict us to the circular arc.

**中文：** 目标函数是多项式，区域闭且有界，所以最大最小都存在。我们把候选点分成三类：内部驻点，直径上的候选点，圆弧上的候选点。圆弧的两个端点也要检查。

**English:** The objective is a polynomial, and the region is closed and bounded, so both absolute extrema exist. Divide candidates into three groups: interior stationary points, candidates on the diameter, and candidates on the circular arc. Check the arc endpoints as well.

**中文：** 先看内部。这里 $y>0$，所以在 $f_y=2y(2-x^2)=0$ 中，可以用 $y>0$ 得到 $x^2=2$。这时 $x$ 非零，再从 $f_x=2x(1-y^2)=0$ 得到 $y=1$。两个点是 $(\pm\sqrt{2},1)$，都在区域内部，函数值是二。

**English:** In the interior, $y>0$. From $f_y=2y(2-x^2)=0$, we can therefore conclude $x^2=2$. Then $x\neq 0$, so $f_x=2x(1-y^2)=0$ gives $y=1$. The two points $(\pm\sqrt{2},1)$ lie inside the region and have function value two.

**【板书 / Board】**

$$f_x=2x(1-y^2),\quad f_y=2y(2-x^2),\qquad (\pm\sqrt{2},1):\ f=2.$$

**中文：** 第二部分是直径。沿 $y=0$，函数变成 $x^2$，参数范围是 $-2\le x\le 2$。所以零点给出值零，两端点给出值四。

**English:** On the diameter $y=0$, the function reduces to $x^2$, with $-2\le x\le 2$. The point $x=0$ gives value zero, and the two endpoints give value four.

**【提问与预期回答 / Question and expected answer】**

**中文：** “横轴上的 $(0,0)$ 在不在区域边界上?” 预期回答：在，它属于上半圆盘的直径边界。

**English:** Is $(0,0)$ on the boundary of this region? Expected answer: yes; it lies on the diameter boundary of the upper half-disc.

**中文：** 第三部分是圆弧。这里 $x^2+y^2=4$，我们观察到函数只涉及 $x^2$ 与 $y^2$，所以可以令 $s=y^2$，从而 $x^2=4-s$，$s$ 的范围是 $[0,4]$。

**English:** On the circular arc, $x^2+y^2=4$. The function involves only $x^2$ and $y^2$, so put $s=y^2$. Then $x^2=4-s$, with $s\in[0,4]$.

**中文：** 这一步把一条弧上的最值问题变成一个闭区间上的一元函数最值问题。代入后得到 $p(s)=s^2-3s+4$，导数是 $2s-3$。所以除了两个端点，还要查 $s=3/2$。

**English:** This turns the arc problem into an ordinary one-variable extremum problem on a closed interval. Substitution gives $p(s)=s^2-3s+4$, with derivative $2s-3$. In addition to the endpoints, check $s=3/2$.

**【板书 / Board】**

$$p(s)=s^2-3s+4,\quad 0\le s\le 4,\quad p'(s)=2s-3.$$

**中文：** 三个参数值 $0,3/2,4$ 对应的函数值是 $4,7/4,8$。注意，$s$ 是 $y^2$，不是 $y$。当 $s=3/2$，我们还原坐标，得到 $y=\sqrt{3/2}$，$x=\pm\sqrt{5/2}$。当 $s=4$，就是最高处的点 $(0,2)$。

**English:** The parameters $0,3/2,4$ give values $4,7/4,8$. Remember that $s=y^2$, not $y$. At $s=3/2$, recover $y=\sqrt{3/2}$ and $x=\pm\sqrt{5/2}$. At $s=4$, we obtain the top point $(0,2)$.

**中文：** 现在可以比较全部候选值了：内部给二；直径给零和四；圆弧给四、四分之七和八。因此唯一最小值是零，在 $(0,0)$；唯一最大值是八，在 $(0,2)$。

**English:** Now compare every candidate value: the interior gives two; the diameter gives zero and four; the arc gives four, seven quarters, and eight. The unique minimum is zero at $(0,0)$, and the unique maximum is eight at $(0,2)$.

**【板书 / Board】**

| 位置 / Location | 候选点 / Candidates | $f$ |
|---|---|---|
| 内点 / Interior | $(\pm\sqrt{2},1)$ | 2 |
| 直径 / Diameter | $(0,0)$ | 0 |
| 弧端点 / Arc endpoints | $(\pm2,0)$ | 4 |
| 弧上驻点 / Arc critical | $(\pm\sqrt{5/2},\sqrt{3/2})$ | 7/4 |
| 弧顶 / Top | $(0,2)$ | 8 |

**中文：** 这题最后一步的比较非常重要。驻点只是候选点，不等于绝对极值；边界上的一元最值也只是整块区域的候选值。必须把所有类别放在一起比较。

**English:** The final comparison is essential. A stationary point is a candidate, not automatically an absolute extremum. Likewise, an extremum of a boundary restriction is only a candidate for the whole region. Compare all groups together.

**中文：** 我们用 $s=y^2$ 的简化，依赖于这道函数只涉及平方。如果下一题有 $xy$、$x$ 或 $y$ 的一次项，就不能直接照搬这个替换而忽略坐标符号。

**English:** The substitution $s = y^2$ works because this function depends only on squares. If another problem contains $xy$ or linear terms in $x$ or $y$, do not copy this substitution blindly and discard the coordinate signs.

**【提问与预期回答 / Question and expected answer】**

**中文：**“如果只检查圆弧，这道题会漏掉哪个最终答案?”预期回答：会漏掉直径上 $(0,0)$ 的最小值零。

**English:** Which final answer would be missed if we checked only the circular arc? Expected answer: the minimum zero at $(0,0)$ on the diameter.

## 175–180 分钟：回顾与答疑 / Recap and questions

**【课堂动作 / Classroom action】**

**中文：**用约 2 分钟回顾，余下约 3 分钟根据现场问题答疑。不重新展开整道长题。

**English:** Use about 2 minutes for the recap and the remaining 3 minutes for questions from the class. Do not restart an entire lengthy solution.

**中文：**今天我们练了十道题。请大家用四句话检查自己有没有掌握方法。

**English:** We have worked through ten problems today. Use four reminders to check whether you have understood the methods.

**中文：**第一，方向导数用单位方向，曲线单位切向量也要除以长度。第二，证明极限不存在可以找不同路径，但几条相同路径不能证明存在。第三，连续和偏导存在仍然不足以保证可微，要检查线性近似的误差除以距离。第四，最值题要保留所有分支，查全边界和端点，最后比较值并写出取值点。

**English:** First, directional derivatives use unit directions, and unit tangent vectors are also normalized. Second, different paths can disprove a limit, but agreement along a few paths does not prove existence. Third, continuity and existence of partial derivatives still do not guarantee differentiability: check the error of the linear approximation divided by distance. Fourth, extremum problems require every branch, every boundary component, and the endpoints; compare the values and state all attainment points.

**中文：**大家现在可以问一个自己最不确定的步骤。比如第七题的某一项从哪里来，第九题为什么要保留零分支，或者第十题怎样确定边界没有漏。

**English:** You can now ask about a step you are unsure of: for example, where a particular term in Problem 7 comes from, why the zero branch is needed in Problem 9, or how to ensure that no boundary component is missed in Problem 10.

**中文：**课后拿到解答时，建议先遮住结果自己重做，再对照第一处不同的步骤。重点找出是方法选错、条件漏掉，还是计算出错。谢谢大家。

**English:** When you receive the solutions after class, try the problems again with the answers covered, then identify the first step where your working differs. Decide whether the issue was the choice of method, an omitted condition, or an arithmetic error. Thank you, everyone.